Hints: Integrals – 4. Series and integrals

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Expand $\frac{1}{1\mp x}$ and $\ln(1+x)$ in powers of x and use $\int_0^1 x^n \ln x\,dx = -\frac{1}{(n+1)^2}$. The first series has constant sign; the other two are alternating (dominated or Abel/Leibniz on $[0,1]$).

Hint 2. You get $-\zeta(2) = -\frac{\pi^2}{6}$, $-\eta(2) = -\frac{\pi^2}{12}$ and $\eta(2) = \sum \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$. The factor $\frac{1}{2}$: $\eta(2) = \zeta(2) - 2\sum \frac{1}{(2n)^2} = \left(1 - \frac{2}{4}\right)\zeta(2)$.

Problem 2

Hint 1. $\frac{\arctan x}{x} = \sum_{k\geqslant 0} \frac{(-1)^k x^{2k}}{2k+1}$; the partial sums converge uniformly on $[0,1]$ (alternating series with remainder at most $\frac{x^{2N}}{2N+1} \leqslant \frac{1}{2N+1}$). Integrate termwise.

Hint 2. For the second, substitute $t = \tan x$: $\int_0^{\pi/4} \ln\tan x\,dx = \int_0^1 \frac{\ln t}{1+t^2}dt = \sum_k (-1)^k \int_0^1 t^{2k}\ln t\,dt = -\sum_k \frac{(-1)^k}{(2k+1)^2} = -G$. (Alternatively, integrate the first integral by parts.)

Problem 3

Hint 1. $I_n = \int_0^1 \frac{x^n}{1+x}dx$ satisfies $I_n + I_{n-1} = \int_0^1 x^{n-1}dx = \frac{1}{n}$ and $I_0 = \ln 2$. So $I_n = (-1)^n\left(\ln 2 - \sum_{k=1}^n \frac{(-1)^{k-1}}{k}\right)$.

Hint 2. I_n is decreasing in n, so $\frac{1}{n+1} = I_n + I_{n+1} \leqslant 2I_n \leqslant I_{n-1} + I_n = \frac{1}{n}$, which gives $I_n \sim \frac{1}{2n}$.

Problem 4

Hint 1. On $(1,\infty)$ write $\frac{1}{1+x^2} = \frac{1}{x^2}\cdot\frac{1}{1+x^{-2}} = \sum_k (-1)^k x^{-2k-2}$, and similarly $\frac{x^{a-1}}{1+x} = \sum_n (-1)^n x^{a-2-n}$. On $(0,1)$ expand in powers of x.

Hint 2. For $\frac{\ln x}{1+x^2}$ the two parts give $-G$ and $+G$, total 0. For $0 < a < 1$, $\int_0^\infty \frac{x^{a-1}}{1+x}dx = \sum_{n\geqslant 0}(-1)^n\left(\frac{1}{n+a} + \frac{1}{n+1-a}\right) = \frac{1}{a} + \sum_{n\geqslant 1}(-1)^n \frac{2a}{a^2-n^2}$.

Hint 3. This is the partial-fraction expansion of the cosecant, $\frac{\pi}{\sin\pi a} = \frac{1}{a} + \sum_{n\geqslant 1}(-1)^n\frac{2a}{a^2-n^2}$, so the answer is $\frac{\pi}{\sin\pi a}$.

Problem 5

Hint 1. $\frac{1}{\binom{2n}{n}} = \frac{(n!)^2}{(2n)!} = \frac{n}{2}B(n,n)$, so $\frac{1}{n^2\binom{2n}{n}} = \frac{1}{2n}\int_0^1 (x(1-x))^{n-1}\,dx$.

Hint 2. Summing under the integral (nonnegative terms): $\sum_{n\geqslant 1}\frac{u^{n-1}}{2n} = -\frac{\ln(1-u)}{2u}$ with $u = x(1-x)$, so the sum is $-\int_0^1 \frac{\ln(1-x+x^2)}{2x(1-x)}dx$. Split $\frac{1}{x(1-x)} = \frac{1}{x} + \frac{1}{1-x}$ and use the symmetry $x \mapsto 1-x$: it equals $-\int_0^1 \frac{\ln(1-x+x^2)}{x}dx$.

Hint 3. $1 - x + x^2 = \frac{1+x^3}{1+x}$, so the integral is $\int_0^1 \frac{\ln(1+x)}{x}dx - \int_0^1 \frac{\ln(1+x^3)}{x}dx = \frac{\pi^2}{12} - \frac{1}{3}\cdot\frac{\pi^2}{12} = \frac{\pi^2}{18}$ (substitute $t = x^3$ in the second).

Problem 6

Hint 1. Substitute $u = 1/x$: $\int_1^\infty \frac{\{u\}}{u^2}du = \sum_{n\geqslant 1}\int_n^{n+1}\frac{u-n}{u^2}du$.

Hint 2. Each term is $\ln\frac{n+1}{n} - \frac{1}{n+1}$; the partial sums are $\ln(N+1) - (H_{N+1}-1) \to 1-\gamma$.

Problem 7

Hint 1. First integral: either directly, $\int_0^1 \ln(1+x)dx - \int_0^1 \ln(1-x)dx = (2\ln 2 - 1) + 1$, or by the series $\ln\frac{1+x}{1-x} = 2\sum_k \frac{x^{2k+1}}{2k+1}$, which gives $2\sum_k \frac{1}{(2k+1)(2k+2)} = 2\ln 2$. Both work; the answer is $2\ln 2$.

Hint 2. Second: no series needed (an expansion in e^x would only be valid for $x<0$ and lead to a sum you then have to recognise). Write $\frac{1+e^x}{1-e^x} = 1 + \frac{2e^x}{1-e^x}$, so the antiderivative is $x - 2\ln|1-e^x| + C$.

Problem 8

Hint 1. The proof is as in the class sheet: expand $x^{\mp x} = e^{\mp x\ln x}$, use $\int_0^1 x^n \ln^n x\,dx = \frac{(-1)^n n!}{(n+1)^{n+1}}$, justify by monotone resp. dominated convergence.

Hint 2. For $\sum n^{-n}$ the tail after N terms is $\sum_{n>N} n^{-n} \leqslant \sum_{n>N}(N+1)^{-n}$, a geometric series about $(N+1)^{-(N+1)}$; for the alternating series the remainder is at most the first omitted term $(N+1)^{-(N+1)}$. Since $7^{-7} \approx 1.2\cdot 10^{-6}$ and $8^{-8} \approx 6\cdot 10^{-8}$, seven terms suffice in both cases (six do not).

Problem 9

Hint 1. $\frac{1}{e^x-1} = \sum_{k\geqslant 1} e^{-kx}$ and $\int_0^\infty e^{-kx}\sin x\,dx = \frac{1}{k^2+1}$; the interchange is justified by domination ($|\sin x| \leqslant x$ and $\int_0^\infty \frac{x}{e^x-1}dx < \infty$).

Hint 2. Use $\pi\cot\pi z = \frac{1}{z} + \sum_{k\geqslant 1}\frac{2z}{z^2-k^2}$ at $z = i$: $\sum_{k\geqslant 1}\frac{1}{k^2+1} = \frac{\pi\coth\pi - 1}{2}$, where $\pi\coth\pi = i\pi\cot(i\pi)$. So the integral is $\frac{\pi\coth\pi-1}{2} \approx 1.0767$.

Problem 10

Hint 1. Assume f is analytic in $\operatorname{Re} z \geqslant 0$ and decays fast enough there. The function $\pi\cot\pi z$ has residue 1 at each integer, so $\sum_{n=0}^N f(n)$ is a contour integral of $\frac{1}{2i}f(z)\cot(\pi z)$ around $[0,N]$, with only half the residue at $z=0$ if the contour passes through 0 along the imaginary axis; this is where $\frac{f(0)}{2}$ comes from.

Hint 2. In the upper half-plane write $\cot\pi z = -i - \frac{2i}{e^{-2\pi i z}-1}$, in the lower one $\cot\pi z = i + \frac{2i}{e^{2\pi i z}-1}$. The constant parts $\mp i$ give, after deforming to the real axis, exactly $\int_0^\infty f(x)\,dx$; the remaining parts decay exponentially away from the real axis, so their contours can be rotated onto the imaginary axis $z = \pm it$, producing $i\int_0^\infty \frac{f(it)-f(-it)}{e^{2\pi t}-1}dt$.

Hint 3. For $f(x) = \frac{1}{(x+a)^2}$, $a>0$: $f(it) - f(-it) = \frac{-4iat}{(a^2+t^2)^2}$, so the formula reads $\psi'(a) = \sum_{n\geqslant 0}\frac{1}{(n+a)^2} = \frac{1}{a} + \frac{1}{2a^2} + \int_0^\infty \frac{4at\,dt}{(a^2+t^2)^2(e^{2\pi t}-1)}$. Check it numerically (e.g. $a=1$ gives $\zeta(2)$), or compare with Binet's formula for $\ln\Gamma$ differentiated twice.